

# KULDEEP SHUKLA

Cybersecurity Professional | Penetration Tester | SOC Analyst

BSc Computing and IT, CCT College Dublin, 2026  
Dundalk, Co. Louth, Ireland | +353 89 494 6668 | kuldeepshuklan@outlook.com  
linkedin.com/in/1daycrew | kuldeepshukla01.github.io

## PROFESSIONAL SUMMARY

Final-year BSc Computing and IT graduate specialising in cybersecurity, penetration testing, network security, and machine learning-based threat detection. Completed Ethical Hacker training through Cisco Networking Academy, plus EC-Council coursework in Deep Web and Cybersecurity and Android Bug Bounty Hunting. Actively practising on TryHackMe and HackThisSite. Experienced with Kali Linux, Nmap, Wireshark, Burp Suite, and Metasploit. Built a production-grade AI-powered Network Intrusion Detection System (NIDS) using Python and XGBoost, classifying 14 attack types on the CICIDS2017 dataset with over 97 percent accuracy. Hands-on with OWASP Top 10, vulnerability assessment, threat analysis, incident response, and CTF challenges. Seeking an entry-level cybersecurity role in SOC analysis, penetration testing, or security engineering.

## CERTIFICATIONS

- Deep Web and Cybersecurity - EC-Council, 2024
- Android Bug Bounty Hunting: Hunt Like a Rat - CodeRed (EC-Council), 2026
- Ethical Hacker (Course Completion) - Cisco Networking Academy, 2026

## TECHNICAL SKILLS

Penetration Testing: Nmap, Metasploit, Gobuster, Burp Suite, Nikto, Hydra, OWASP Top 10, Bug Bounty Methodology

Network and Packet Analysis: Wireshark, NFStream, TCP/IP, DNS, HTTP/S, Firewalls, IDS, NIDS, Threat Analysis

Programming and Development: Python (scikit-learn, XGBoost, SHAP, Flask, Streamlit), Bash, Java, JavaScript, Git, Docker

Operating Systems: Kali Linux, Debian, Ubuntu, Windows 10/11, macOS, Android OS

Platforms and Labs: TryHackMe, HackThisSite, CTF Challenges, OSINT Tools

Core Competencies: Penetration Testing, Network Security, Ethical Hacking, Vulnerability Assessment, Threat Analysis, SOC, NIDS, IDS, OWASP Top 10, Android Security, Machine Learning, Incident Response, Bug Bounty Methodology, Python and Bash Scripting

## TECHNICAL PROJECTS

### 1DayCrew AI - ML Network Intrusion Detection System (NIDS)

*Python, XGBoost, SHAP, NFStream, Flask, Streamlit, CICIDS2017*

- Built end-to-end ML NIDS in Python using XGBoost trained on CICIDS2017 dataset, classifying 14 attack types (DDoS, PortScan, Brute Force, Web Attacks) with over 97 percent accuracy
- Integrated NFStreamer for real-time packet flow capture and live threat classification with dynamic confidence thresholds
- Developed dual-mode dashboard for live network sniffing and offline PCAP/CSV forensic analysis
- Implemented SHAP TreeExplainer for full prediction explainability, surfacing top 6 driving features per prediction
- Deployed Flask web UI with Claude API integration and ngrok remote access tunnel for secure external connectivity
- Applied SMOTE oversampling and StratifiedKFold cross-validation to address class imbalance across rare attack types

### 1DayCrew OS - Custom Debian-Based Security Distro

*Debian Linux, KDE Plasma, Ollama, Open Interpreter, Plymouth*

- Built custom Debian OS on a 2012 MacBook Pro with KDE Plasma desktop and hardware-accelerated Ollama LLM engine
- Configured Broadcom BCM4331 WiFi drivers (b43 module) and Open Interpreter for AI-driven system control
- Designed Plymouth matrix-rain boot animation and Flask web UI with ngrok remote access agent

### Post-Quantum Cryptography Research - CCT College Dublin

*PQC, Homomorphic Encryption, Zero-Knowledge Proofs, Porter's Five Forces*

- Researched and presented on Post-Quantum Cryptography, Homomorphic Encryption, and Zero-Knowledge Proofs; analysed real-world adoption using Porter's Five Forces framework with JPMorgan Chase as an industry case study

## WORK EXPERIENCE

### Retail Sales Assistant - Calendar Club (Seasonal), 2019 - Present

- Customer-facing sales, payment processing, and stock management in a high-traffic retail environment
- Developed communication, upselling, and problem-solving skills across multiple seasonal contracts

### Kitchen Porter - Rocksalt Cafe, Dundalk, 2024 - Present

- Maintained high performance in a fast-paced team environment, reliability, multitasking, and accountability under pressure

## EDUCATION

### BSc (Hons) Computing and IT - CCT College Dublin, 2022 - 2026

Focus: Cybersecurity, Machine Learning, Network Security, Data Analytics

### Diploma in Information Technology - Hewett Polytechnic, India, Completed June 2020

## LANGUAGES AND ADDITIONAL INFORMATION

Languages: English (Fluent), Hindi (Native). Stamp 2 Work Authorisation (Ireland). Full Clean Irish Driving Licence.